Municipal Revenue collection system

PATNA WOMEN'S COLLEGE

(AUTONOMOUS)

4th Cycle NAAC Accredited at A++ Grade with CGPA 3.51/4

"College with potential for excellence" (CPE) status accord by UGC "PROJECT SYNOPSIS

Municipal Revenue collection system

Submitted to

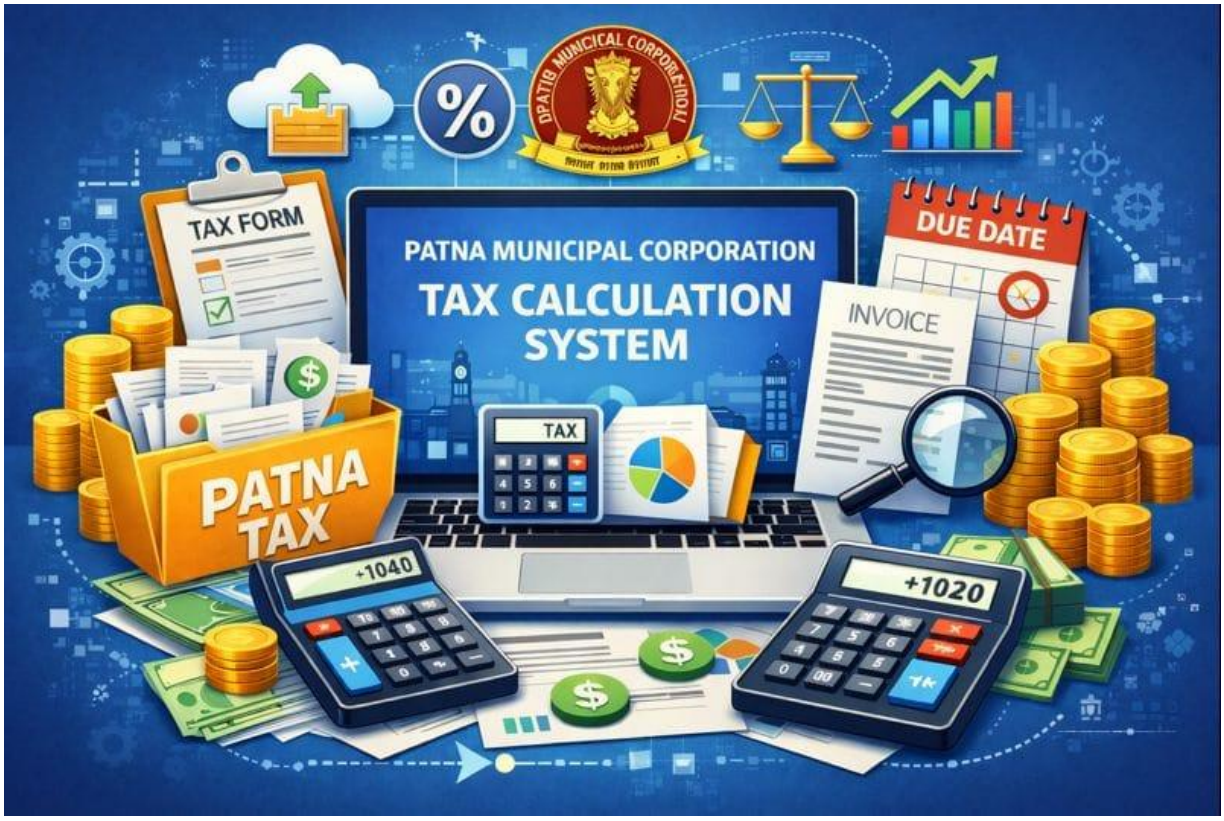
Department of Computer Application (MCA)

In Partial fulfilment of the Requirement from the award of the degree.

Municipal Revenue collection system

PROJECT TITLE

"SYNOPSIS OF Municipal Revenue collection system "



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ABSTRACT

The Patna Municipal Corporation Municipal Revenue collection system
is a computerized system developed to simplify and automate the calculation of municipal taxes such as property tax, water tax, and holding tax. The system collects basic property details including location, property type, area, construction type, and number of floors and uses predefined tax rules to calculate the correct tax amount automatically.

This system reduces manual work and calculation errors, making the tax process faster, more accurate, and transparent. It allows citizens to view their tax details and make payments online, saving time and effort. For the municipal corporation, the system ensures efficient revenue collection, proper record management, and better administrative control.

Overall, the Patna Municipal Corporation Tax Calculation System improves the quality of municipal services by providing a simple, reliable, and user-friendly approach to tax management.

INTRODUCTION

The Patna Municipal Corporation (PMC) Municipal Revenue collection system is a digital platform designed to calculate municipal taxes such as property tax, water tax, and holding tax in an accurate, fast, and transparent manner. The main objective of this system is to reduce manual errors and allow citizens to check and pay their taxes online from home.

In this system, property-related details such as property area, location (zone/ward), property type (residential or commercial), construction type, and floor details are entered. Based on these inputs, the system applies pre-defined tax rules and rates to automatically calculate the payable tax amount.

The Patna Municipal Corporation Tax Calculation System is user-friendly, reliable, and time-saving, benefiting both citizens and municipal authorities. It improves transparency in tax assessment and helps in better revenue management and efficient administration for the city.

OBJECTIVES

The main objective of the Patna Municipal Corporation Tax Calculation System is to automate and simplify the process of calculating municipal taxes such as property tax, water tax, and holding tax. The system aims to provide accurate and error-free tax calculations by using predefined rules and rates. It is designed to be user-friendly so that citizens can easily check their tax details and make online payments without visiting municipal offices. Another objective is to improve transparency and fairness in tax assessment while reducing time, effort, and manual workload for municipal staff. The system also helps in maintaining proper digital records and supports efficient revenue collection and better administration of the city.

Another important objective of the system is to provide convenience to citizens by allowing them to view their property details, check tax amounts, and make payments online without visiting municipal offices. This saves time and effort for the public and improves overall user experience. For the municipal corporation, the system helps in efficient record management, easy tracking of tax payments, and generation of reports for better decision-making. Overall, the objective of this system is to support effective revenue collection, improve administrative efficiency, and contribute to better urban governance in Patna.

H/W & S/W REQUIREMENT

H/W REQUIREMENT

- Processor- Intel Core i5
- RAM – 8GB
- HDD/SDD – 512GB
- Monitor
- Keyword
- Mouse

S/W REQUIREMENT

- OS -Windows 11
- Browser: Google Chrome

Tools and Techniques

- **Frontend** – HTML, CSS, JAVASCRIPT
- **Backend –** NodeJs
- **Database** - MongoDB

Tools and Technologies

HTML- **HTML** stands for **Hyper Text Markup Language**. It is the standard language used to create and structure content on the web. It provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes, and other items. The elements are delineated by tags, written using angle brackets.

CSS- CSS (Cascading Style Sheets) is a language designed to simplify the process of making web pages presentable. It consists of style rules that are interpreted by the browser and applied to the corresponding elements. A style rule set includes a selector and a declaration block. It is used to design the website.

JavaScript- JavaScript is a versatile, dynamically typed programming language used for interactive web applications, supporting both client-side and server-side development, and integrating seamlessly with HTML , CSS and a rich standard library. It is used for validation. It is a single threaded language that executes one task at a time. It executes the code line by line.

Node.js is a JavaScript runtime environment used for server-side programming. It is built on Google Chrome's V8 JavaScript engine, which makes it fast and efficient. Node.js works on a non-blocking, asynchronous model, allowing it to handle multiple requests at the same time using a single thread. It is commonly used to build web servers, REST APIs, and real-time applications such as chat apps and online streaming platforms. Node.js also includes npm (Node Package Manager), which helps developers easily install and manage external libraries. With Node.js, JavaScript can run outside the browser, and because it is cross-platform, it works on Windows, Linux, and macOS.

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MongoDB is a **NoSQL database** used to store data in a flexible and scalable way. It stores data in the form of **documents** using **JSON-like (BSON)** format instead of tables and rows. MongoDB is schema-less, which means data structure can be changed easily without affecting existing data. It is very fast and suitable for handling **large amounts of data** and **real-time applications**. MongoDB supports features like indexing, aggregation, and replication for better performance and reliability. It is commonly used with Node.js to build modern web applications.

System Analysis

System Analysis refers to understanding and documenting the system's requirements. It is a problem-solving technique that improves the system and ensures that all the components work efficiently to accomplish the purpose. It is the process of planning a new system to either replace or complement an existing system

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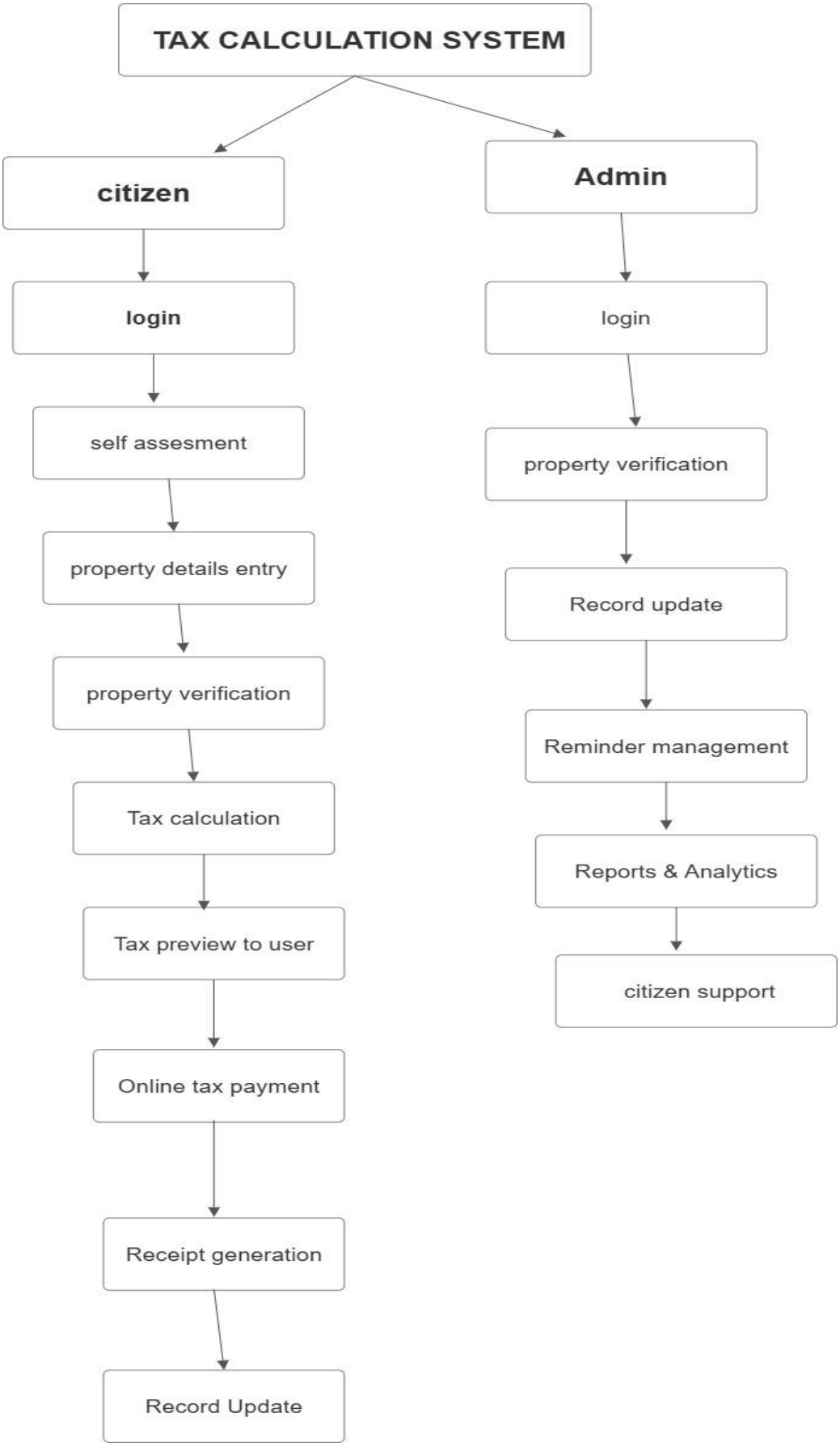
Problem Definition

To address the challenge of providing a seamless, secure, and user-friendly online platform for sharing technical knowledge, where users can create blog posts, interact with others, and stay updated with the latest technology trends. The goal is to build a collaborative environment for learners, professionals, and enthusiasts to share insights and inspire innovation.

Requirement Specification

A **Software Requirement Specification (SRS)** is a complete description of the behaviour of the system to be developed. It includes a set of use cases that describe all the interactions the users will have with the software. Use cases are also known as **Functional Requirements**.

Application Workflow



System Design

System Design is the process of designing the architecture , components , and interfaces for a system so that it meets the end-user requirements . It is also about ensuring the system can withstand real-world conditions . A well-designed system is not only reliable and scalable but also easy to maintain and evolve as requirement change .

In this phase we discuss about the following:

1) Data Flow Diagram
2) Entity-Relationship Diagram
3) Unified Modeling Language

1. Data Flow Diagram

A DATA FLOW DIAGRAM (DFD) is a graphical representation of the flow of the data within a system. It is commonly used in software engineering and systems. In a DFD, processes, data stores, data flows, and external entities are represented by symbols, and the connections between them show how data moves through the system.

The main components of DFD are as follows :

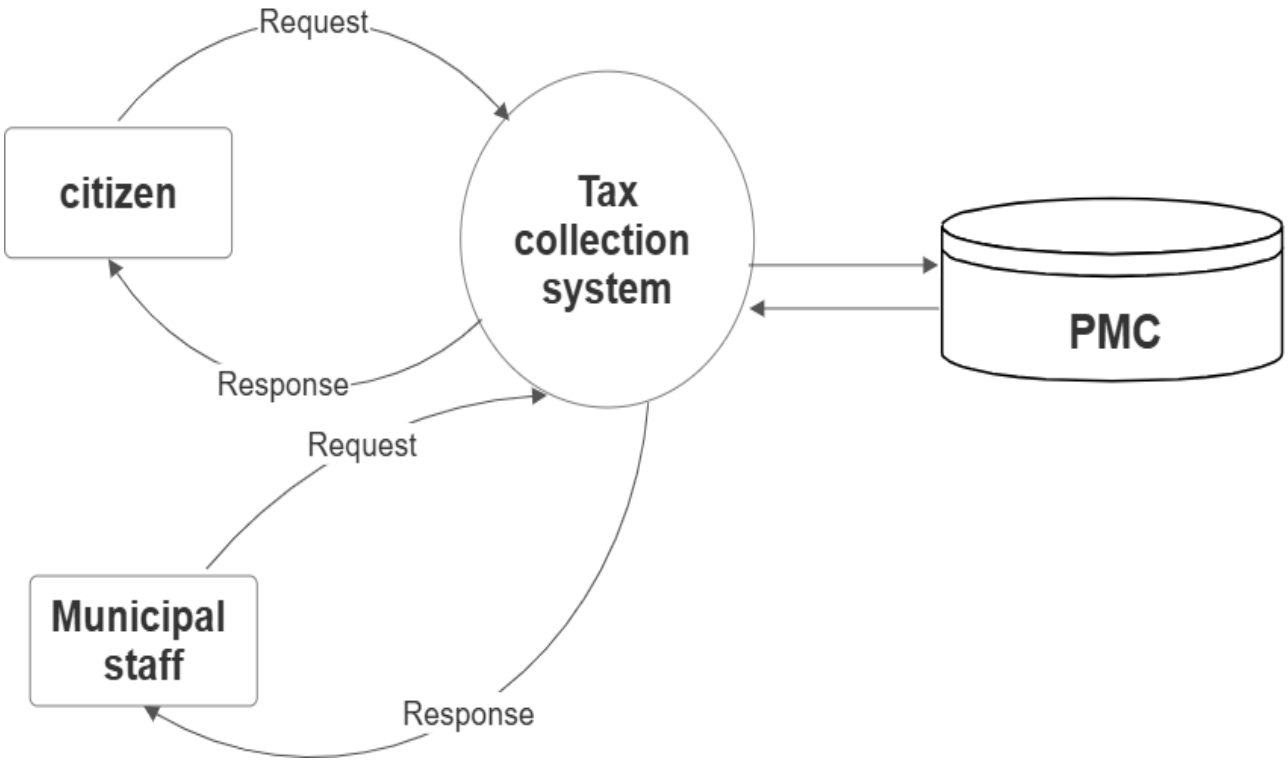
Processes: These are the functions or activities that transform data within the system. They're represented by circles or rectangles in a DFD.

Data Flows: These are the pathways through which data moves from one part of the system to another. They're represented by arrows and show the direction of data flow.

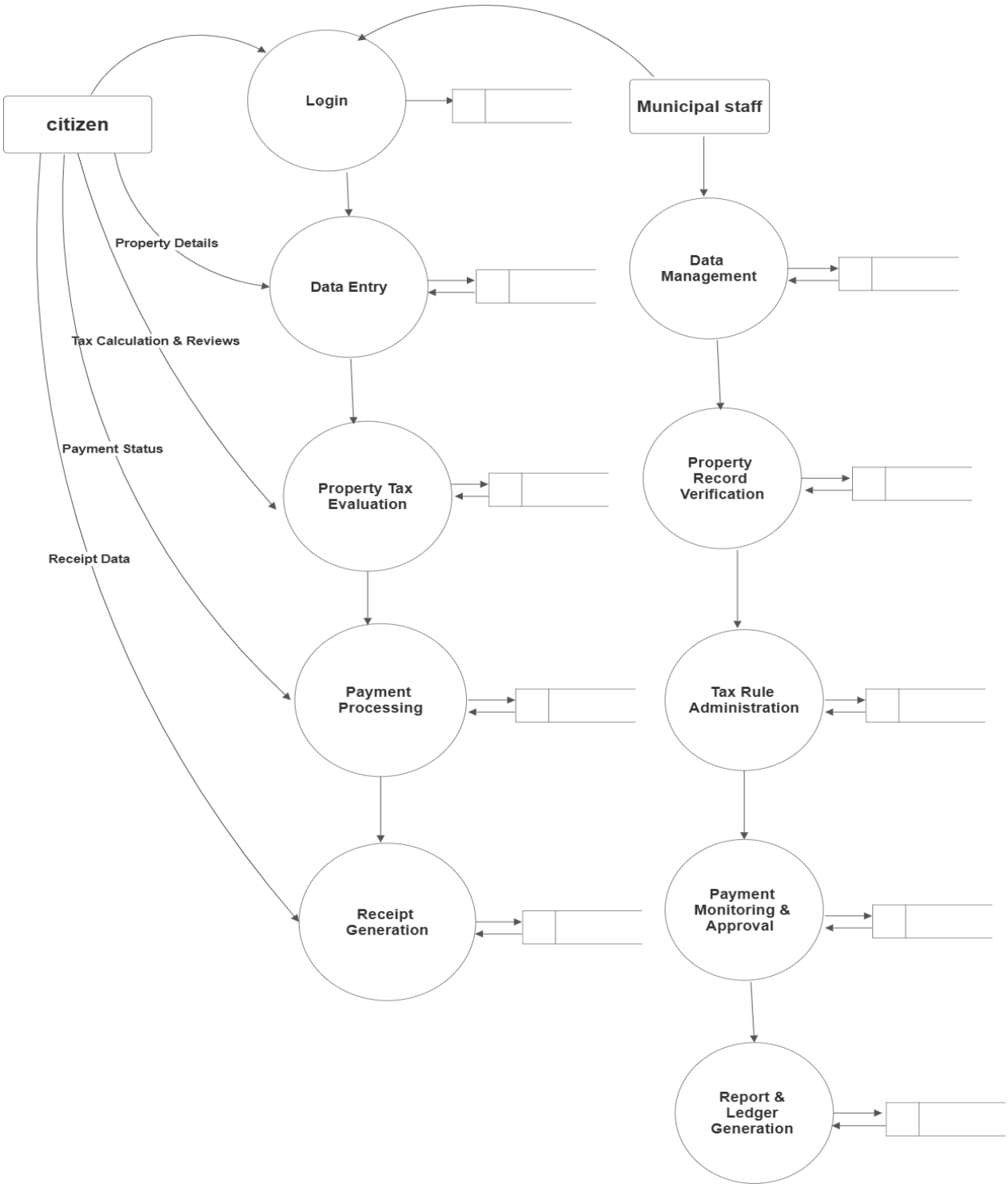
Data Stores: These are repositories where data is stored within the system. They're represented by rectangles with two parallel lines on one side.

External Entities: These are sources or destinations of data that are outside of the system is being analysed. They can be users, other systems, or organizations interacting with the system under study. They're represented by squares or rectangles.

0-Level DFD

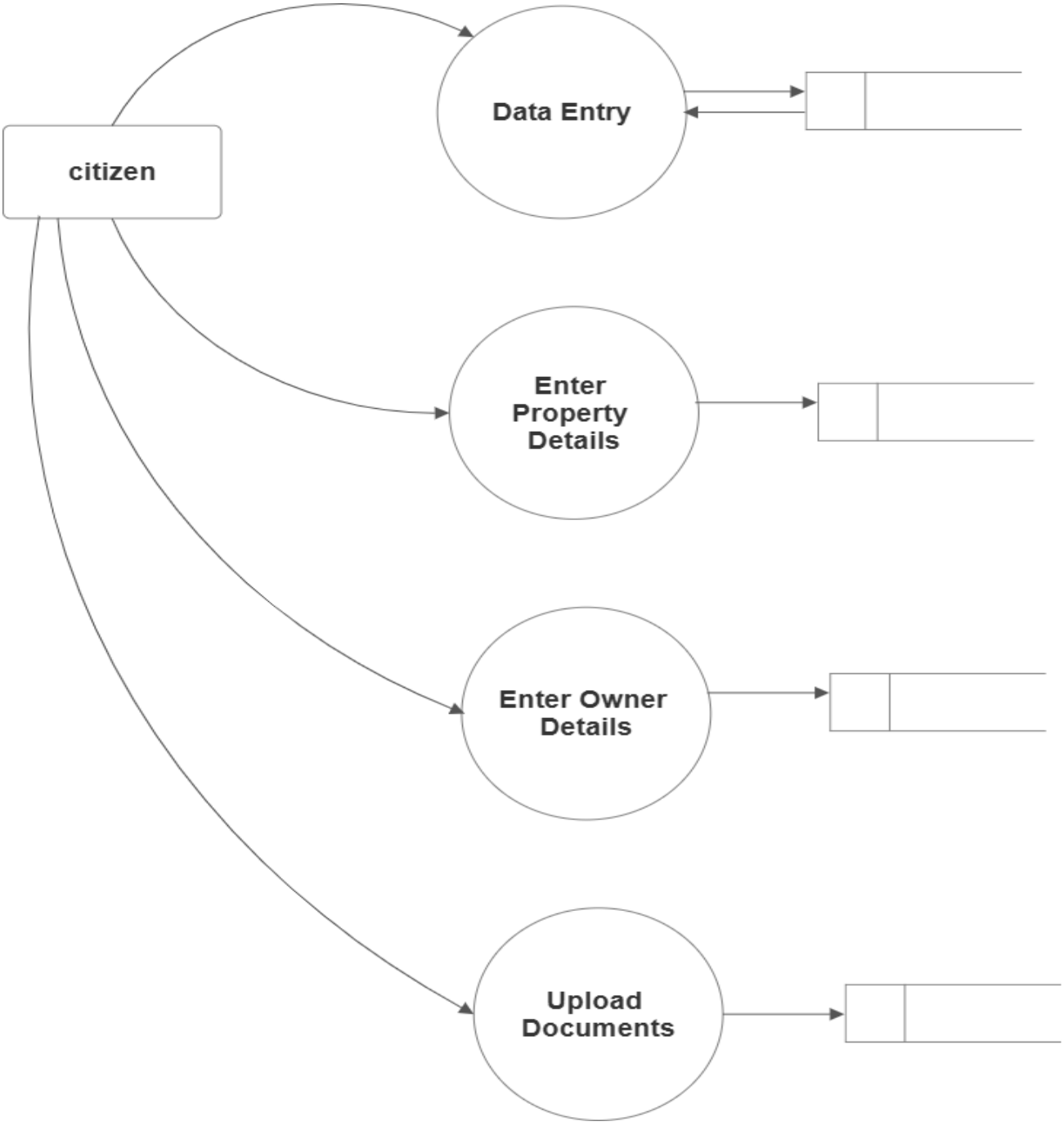

Request
citizen
Tax
collection
system
PMC
Response
Request
Municipal
staff
Response

1st-Level DFD

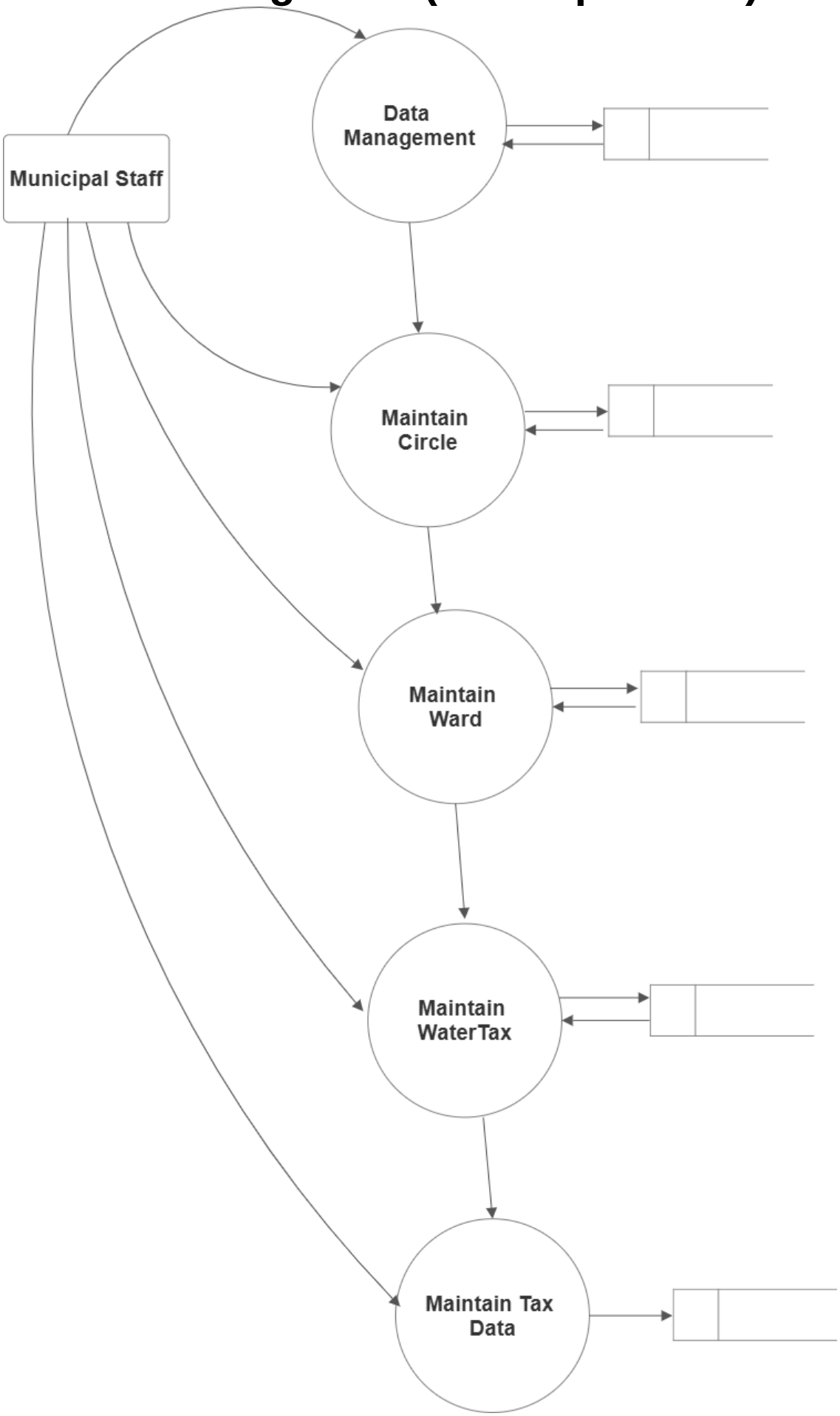


2nd-Level DFD

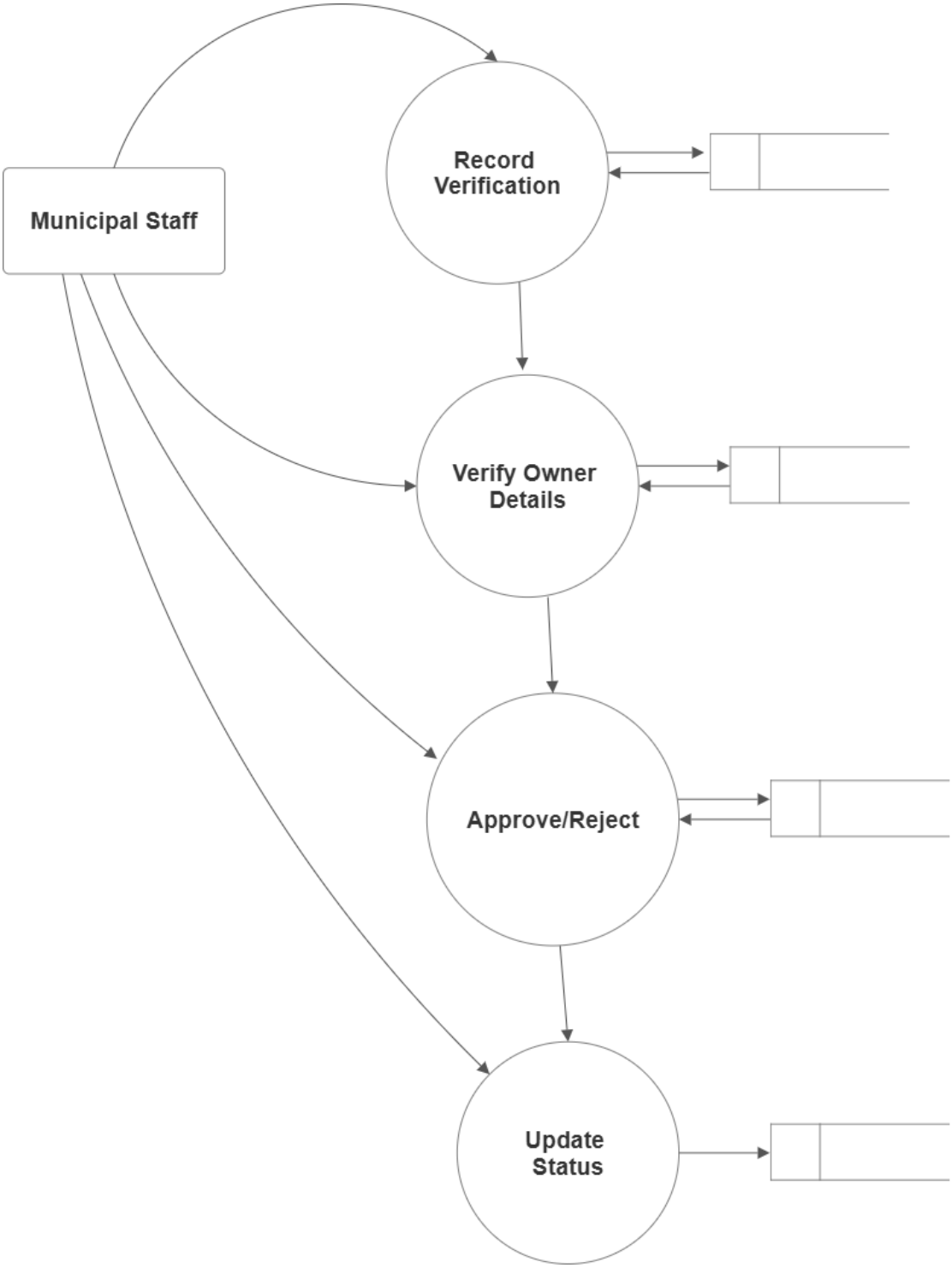
1.Data Entry(citizen)



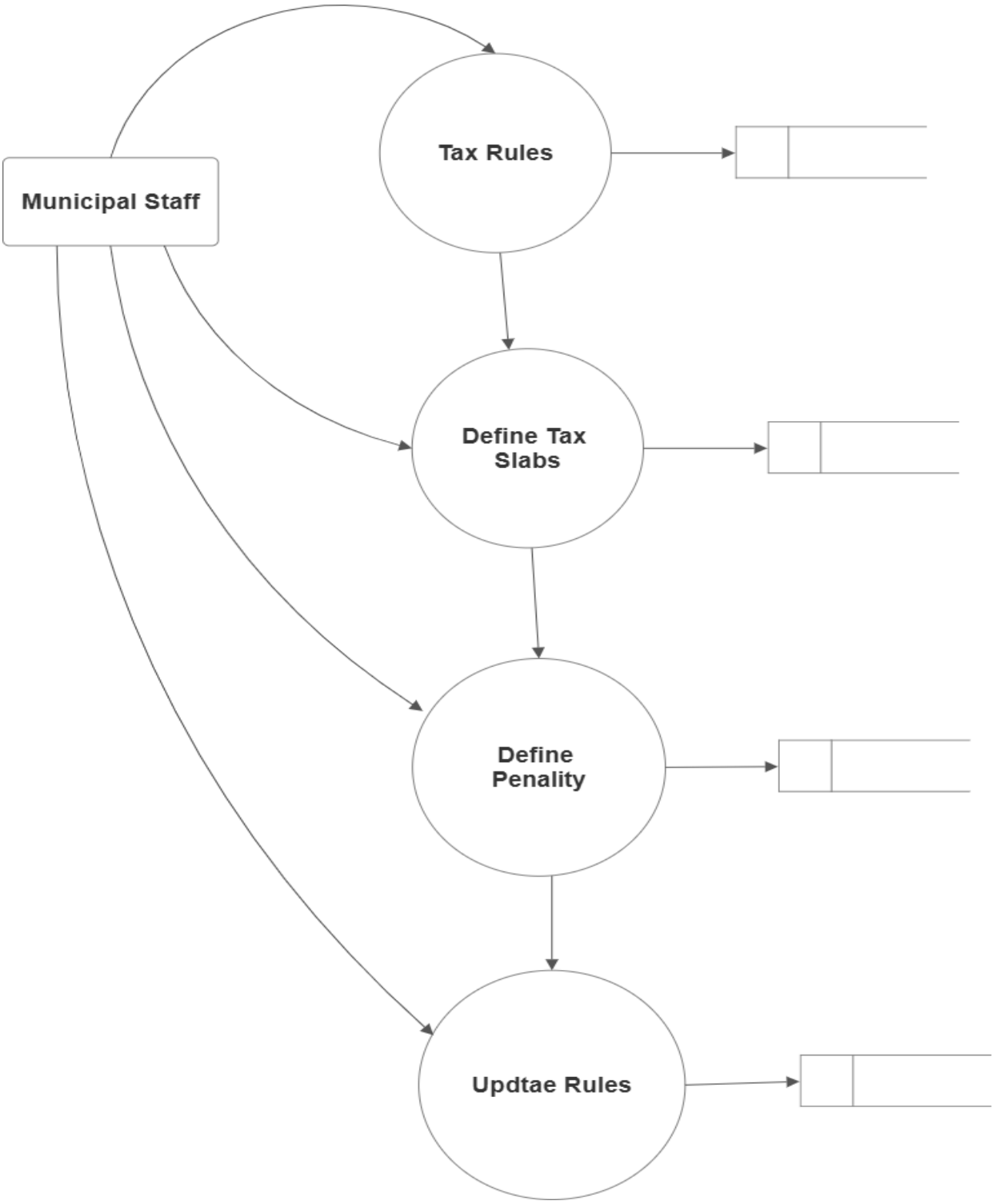
2. Data Management(Municipal Staff)



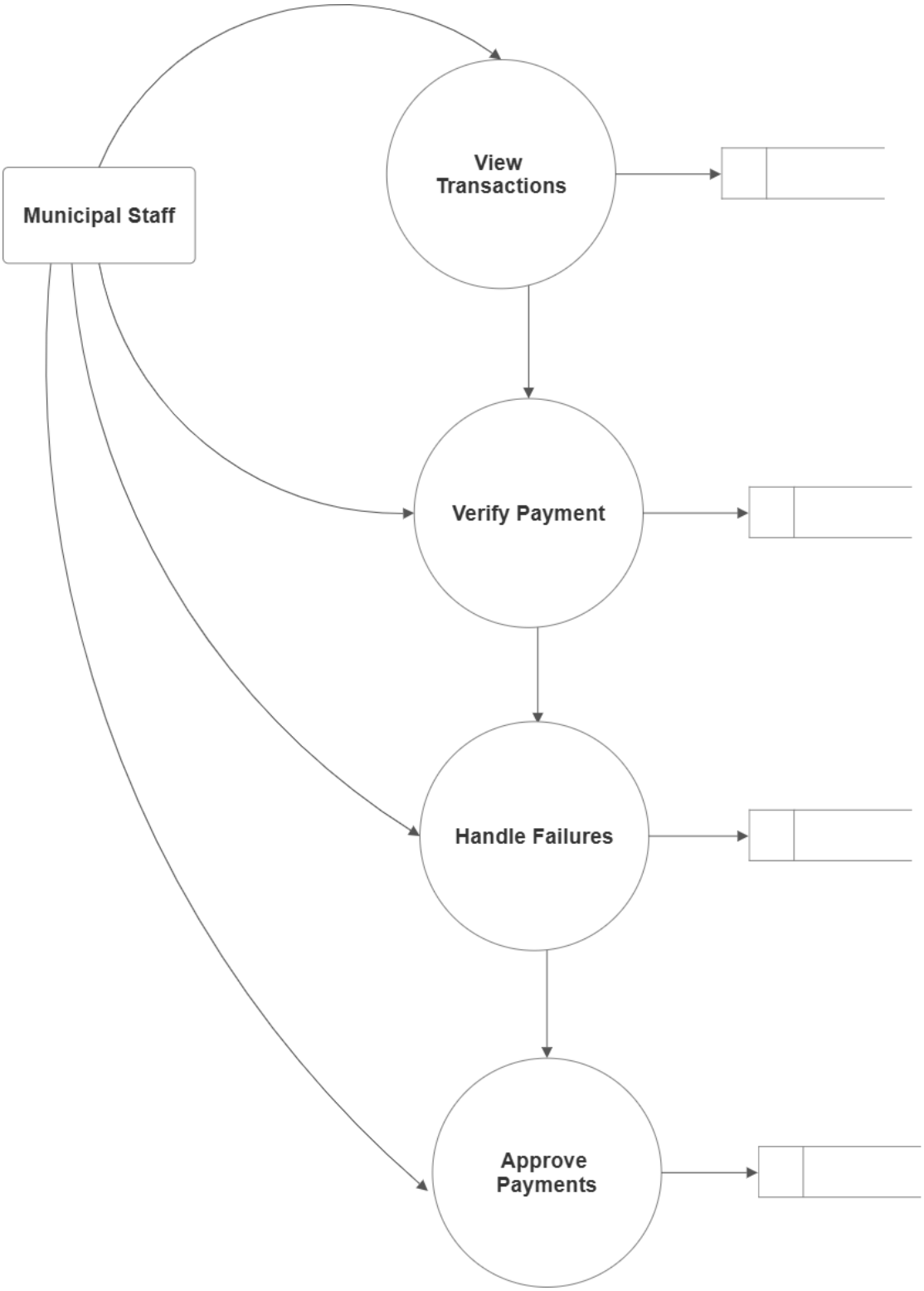
3.Record Verification



4.Tax Rule


Municipal Staff
Tax Rules
Define Tax Slabs
Define Penality
Updtae Rules

5.Payment Monitoring



3.UML(Unified Modling Language

(Unified Modelling Language) is a standardized visual language used in software engineering to describe, specify, design, and document the structure and behaviour of software systems. It provides a set of graphical notations that help developers and stakeholders communicate complex system designs.

TYPES OF UML DIAGRAM

UML includes several types of diagrams, which can be categorized into two main groups:

1.Structural Diagrams: Represent the static aspects of the system, such as its components and their relationships.

a.) Class Diagram: It Shows classes, their attributes, methods, and relationships.

b.) Object Diagram: Depicts instances of classes and their relationships at a particular point in time.

c.) Component Diagram: Displays the components and their dependencies in a system.

d.) Deployment Diagram: Represents the physical deployment of artifacts on hardware components.

2. Behavioural Diagrams: Focus on the dynamic aspects of the system, such as how it behaves and interacts over time.

a.) Use Case Diagram: Illustrates the system's functionality from the user's perspective.

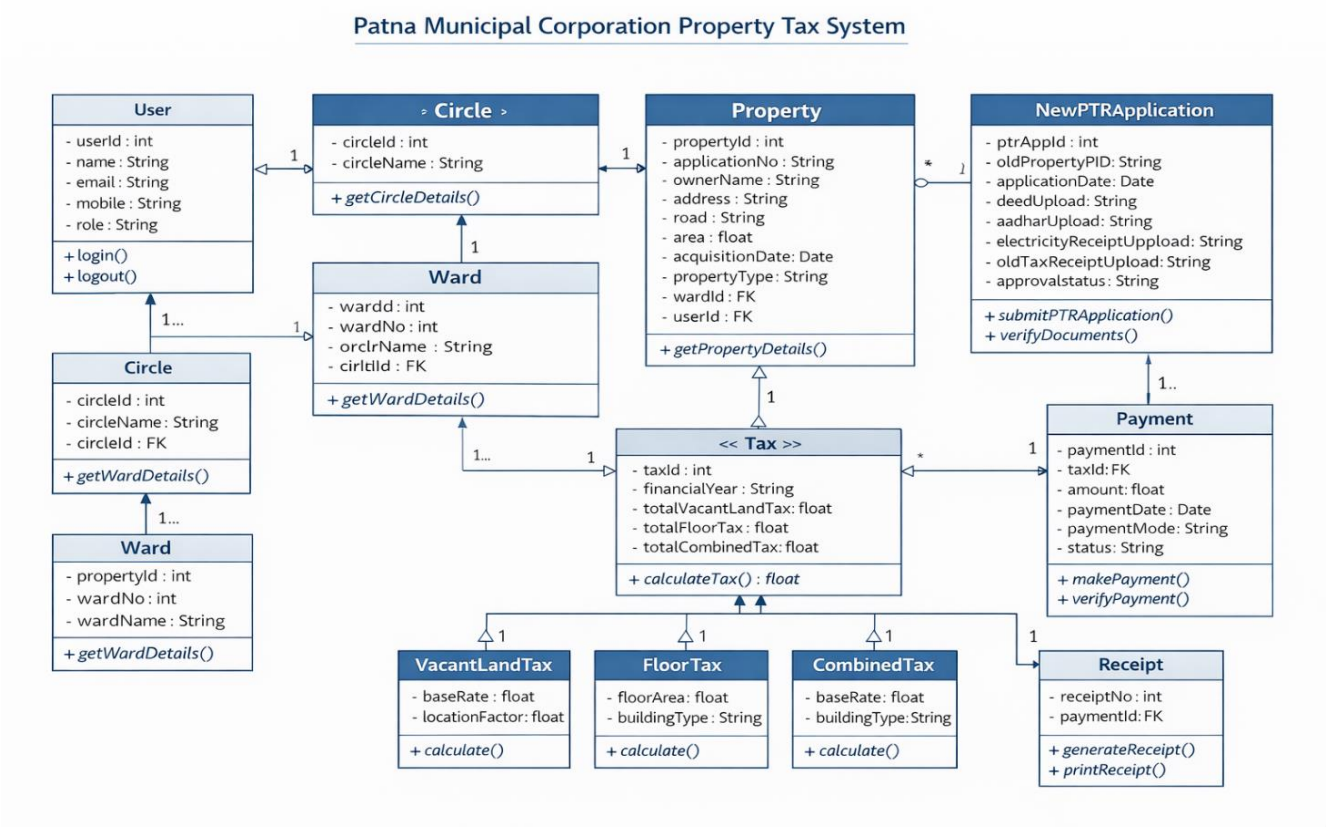
b.) Sequence Diagram: Depicts the sequence of interactions between objects or components over time.

c.) Activity Diagram: Shows the flow of activities or operations within the system.

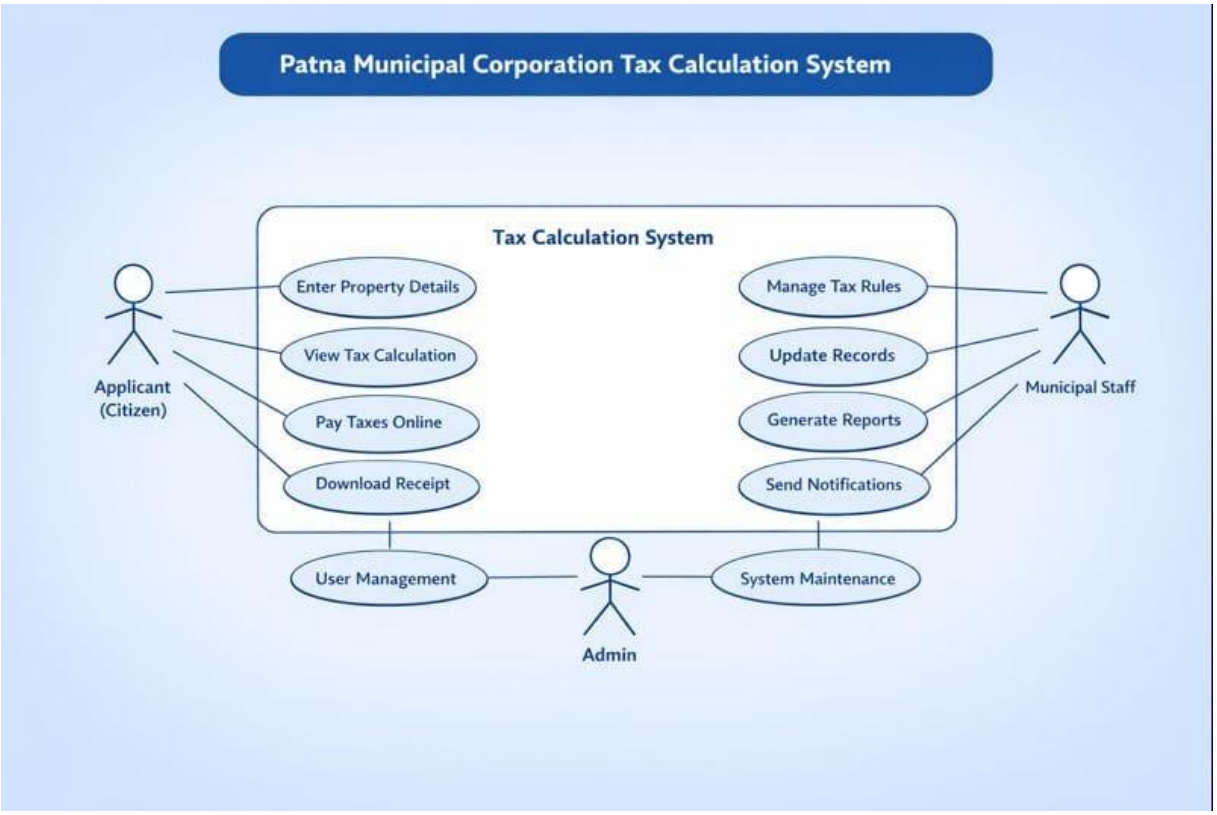
d.) State Diagram: Models the states an object can be in and the transitions between those states.

3.UML Diagram

Class Diagram



2.Use Case Diagram



2.Entity Relationship Diagram

The Entity Relational Model is a model for identifying entities to be represented in the database and representation of how those entities are related. The ER data model specifies enterprise schema that represents the overall logical structure of a database graphically.

The Entity Relationship Diagram explains the relationship among the entities present in the database. ER models are used to model real-world objects like a person, a car, or a company and the relation between these real-world objects. In short, the ER Diagram is the structural format of the database.

Roles of ERD :

- *Database Design* : It is mainly used to model the database structure for a system.it define what tables will be present in the database, the attributes of those tables, and how they are connected, making it easier to design and understand the relationship between various entities like customers, products, etc.
- *Defining Relationships* : ERDs define the relationship between entities in a database. These can be: **one-to-one, one-to-many, Many-to-Many , Many-to One**.

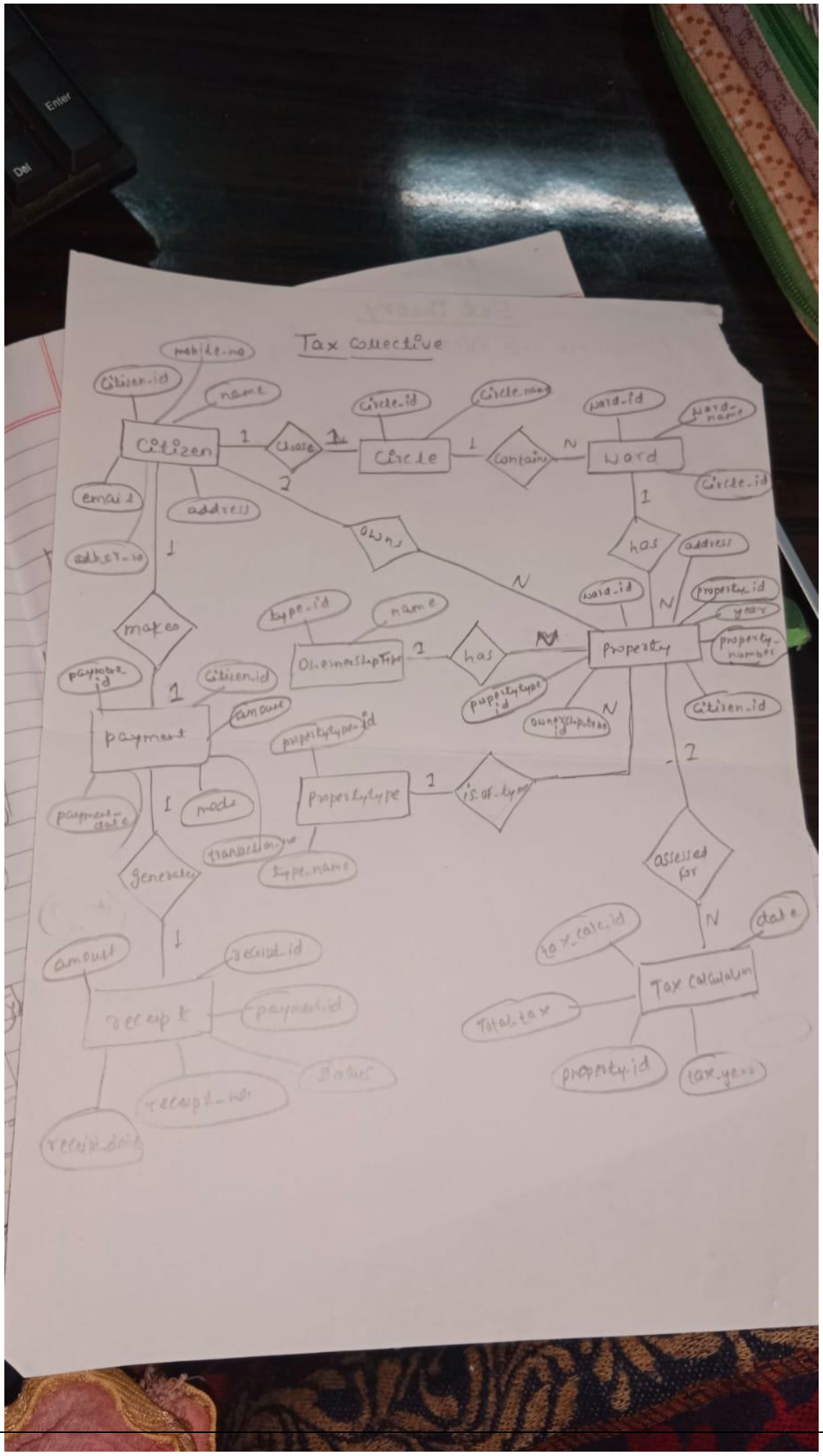
Data Integrity and Normalization : ERDs helps in maintain data integrity by setting clear relationships and constraints between entities. They also helps in **normalization,** which reduces redundancy and ensure the database is well structured.

Symbols used in ER Diagram

ER Model is used to model the logical view of the system from a data perspective which consists of these symbols:

- **Rectangles:** Rectangles represent Entities in the ER Model.
- **Ellipses:** Ellipses represent Attributes in the ER Model.
- **Diamond:** Diamonds represent Relationships among Entities.
- **Lines:** Lines represent attributes to entities and entity sets with other relationship types.
- **Double Ellipse:** Double Ellipses represent the multi-valued attributes
- **Double Rectangle:** Double Rectangle represents a Weak Entity.

Entity Relationship Diagram


Tax collective
Citizen
Circle
Ward
Property
Ownership type
Property type
Payment
Receipt
Tax calculation
Citizen_id
name
mobile_no
email
address
adhar_no
Circle_id
Circle_name
Ward_id
Ward_name
Property_id
year
property_number
type_id
payment_id
amount
mode
payment_date
receipt_id
receipt_no
receipt_date
tax_calc_id
date
Total_tax
tax_year
makes
generates
owns
has
contains
assessed for
is_of_type

Testing

Testing is essential to ensure that the Tax Calculation System works accurately, securely, and efficiently while meeting legal, user, and business requirements. It helps verify correct tax computation, data security, performance, and overall system reliability.

Types of Testing

1. Functional Testing

Functional testing ensures that all features of the tax calculation system work correctly according to defined requirements and tax rules.

Examples:

- **User Login:** Can users log in using valid credentials?
- **Income Entry:** Can users enter salary, business income, and other income details correctly?
- **Tax Calculation:** Does the system calculate tax accurately based on income slabs, deductions, and exemptions?
- **Deductions:** Are deductions (e.g., Section 80C, 80D) applied correctly?
- **Tax Report Generation:** Can users generate and download tax calculation reports without errors?

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2. Non-Functional Testing

Non-functional testing evaluates system performance, security, usability, and compatibility under different conditions.

Examples:

- **Performance:** Can the system handle multiple users calculating taxes simultaneously without slowing down?
- **Security:** Is sensitive financial data protected from unauthorized access and common attacks (e.g., SQL injection, data leakage)?
- **Usability:** Is the tax calculation process easy to understand for non-technical users?
- **Compatibility:** Does the system work correctly on different browsers (Chrome, Firefox, Edge) and devices (desktop, tablet, mobile)?

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3. Regression Testing

Regression testing ensures that new updates or changes do not affect existing system functionality.

Examples:

- After adding a new tax slab, does the old tax calculation logic still work correctly?
- After updating deduction rules, can users still log in and generate tax reports without issues?
- After fixing a calculation bug, does the system still store user data correctly?

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4. User Acceptance Testing (UAT)

User acceptance testing confirms that the system meets user expectations and legal requirements before final deployment.

Examples:

- Can a user easily enter income details and calculate tax without confusion?
- Does the generated tax summary clearly show payable tax or refund amount?
- Can an admin easily update tax slabs, deduction limits, and financial year settings?

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5. Integration Testing

Integration testing ensures that all system modules work together seamlessly.

Examples:

- Does the income entry module correctly integrate with the tax calculation engine?
- When a user updates income details, does the tax amount update automatically?
- Does report generation integrate properly with the database to fetch correct calculation data?

Future Enhancement

Future Enhancements for Patna Municipal Corporation Tax Calculation System

1. Al-Based Property Valuation

Artificial Intelligence can be used to automatically estimate themarket value of properties, reducing manual errors and under-valuation.

2. Blockchain for Secure Tax Records

Blockchain technology can ensure tamper-proof, transparent, and highly secure storage of tax and property records.

3. Intelligent Chatbot & Voice Assistance

An Al-powered chatbot or voice assistant can help citizens calculate tax, check dues, and get instant support without visiting offices.

4. Predictive Revenue Analytics

Machine learning models can be used to forecast future tax revenue, helping authorities in better financial planning.

5. Smart Defaulter Identification System

Advanced analytics can identify chronic defaulters and automatically generate notices or reminders.

6. Smart City System Integration

Integration with smart meters, building approval systems, and utility da ases can enable real-time property updates.

Conclusion

The Patna Municipal Corporation Tax Calculation System can be significantly improved by adopting modern technologies such as Artificial Intelligence, Blockchain, and advanced data analytics. These future enhancements will increase accuracy, transparency, and efficiency in property tax assessment and collection. Features like AI-based valuation, smart defaulter identification, predictive revenue analytics, and intelligent citizen support systems will reduce manual work, minimize errors, and enhance user satisfaction. Overall, the upgraded system will support better governance, increase municipal revenue, and contribute effectively to the development of a smart and digitally empowered Patna city.

Reference

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